Toward Ludovico: Student Preferences between Light- and Dark-Mode PowerPoints

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Abstract

Apparently studying whether students prefer light mode or dark mode creates very polarizing reviews. Students prefer dark mode; reviewers are more polarized.

1 Introduction

Lecture Materials such as PowerPoint presentations, slide decks, and other shared course materials are a core learning artifact provided by instructors to assist students during their studies. However, these materials often use default settings from the applications used to create them, colloquially referred to as 'Light Mode'. This version presents lecture or application content using a white or light background superimposed with black or dark text. In recent years, developers and other occupations that rely on computers have begun to incorporate a 'Dark Mode' color scheme. Although this difference may appear like a small quality-of-life feature, application users have increasingly begun to request 'Dark Mode' themes as an option, and there has yet to be research exploring the preference among student populations.

As part of this work, the author presents an experience report of their implementation of course materials using a 'Dark Mode' color scheme in comparison to a 'Light Mode' color scheme. Students were recruited from an introductory CS1 course and an upper-level AI course (CS4), over two semesters. In total, 311 students consented to have their course clickstream data analyzed. Students were presented with the option to view the day's lecture materials using either the 'light' or 'dark' color scheme. Although most of the students showed

Color	Light Mode	Example	Dark Mode	Example
White	#ECF0F1		#ECF0F1	
Black	#272822		#272822	
Red	#990000		#FF2E2E	
Orange	#D64700		#FF884D	
Yellow	#FAD000		#FFE563	
Green	#7D8C1F		#D7EF43	
Blue	#427E93		#71BFDA	
Indigo	#2A49B7		#748BDF	

Table 1: Hex code values for common colors based on University Branding. Put your Communications Department to work if your institution does not have one.

a clear preference for the 'dark mode' color scheme, a small but notable portion preferred the 'light mode' scheme. A comparison of course performance between light and dark mode users was unable to identify differences between the cohorts. Finally, we summarize students' end-of-semester survey responses on their perceptions of Dark Mode lecture materials.

This is the author's first SIGBOVIK submission, so here's what I wanted to know:

RQ1 What color scheme PowerPoints did students prefer?

RQ2 Did that preference change during the semester?

RQ3 Did that preference impact their overall course performance?

2 Background

2.1 Dark Mode

Pendersen et al. describe 'dark mode' as "an inverted text background-foreground configuration where the text in the foreground is bright and the background is dark, whereas light mode involves dark text in the foreground on a bright background" [10]. While Pendersen et al. found no significance between light and dark mode preference in regards to productivity, our study focuses on student preference between these two color schemes for lecture materials and its potential impact on their academic pursuits.

Table 1 provides the hex code values that we used for both our Light Mode and Dark Mode versions. Light Mode's accent colors were developed by the author's university as part of a larger branding initiative. The Light Mode color scheme was designed for publishing purposes, which may have different visualization purposes compared to lecture materials. We addressed color contrast issues for Dark Mode by lightening the Light Mode accent colors by 40% (calculated via Microsoft PowerPoint).

2.2 Eye Strain and Screen Fatigue

There has been extensive research on the impacts of blue light in relation to sleep, circadian rhythms, and workplace performance [9, 1, 14, 5]. Hatori et al discussed growing concerns over potential health hazards produced from blue light with regard to disruptions in human sleep cycles [6]. Likewise, numerous studies have explored the students' sleep habits and their correlation to academic performance, covering performance at the primary school-level [3] to the elderly [4]. Further research and advocacy has expressed concerns over general screen time and impacts impact on the viewer's mental health [13].

2.3 Color Theory and Multimedia Learning

Sweller used worked examples in geometry to explore differing dimensions of cognitive load and split-attention theory at their relation with an individual's working memory, known as Cognitive Load Theory [12]. Mayer further expanded on the cognitive load imposed by split-attention effects in multimedia learning [8]. We use Sweller and Mayer's research into the multimedia elements of learning materials as motivations for our own research into students' color scheme selection. The color scheme variable presents a potential germane resource that instructors could implement with their course material to help reduce students' overall cognitive load. While we do not explicitly quantify our results to confirm or refute cognitive load theory, we present the frequency and potential effects of these educational multimedia elements.

3 Study Design

Students enrolled in six 16-week courses across two semesters were recruited to take part in this study. Two courses were a 100-level introductory CS1 course and four were a 400-level advanced CS course covering AI concepts (CS4). The instructor introduced the study to students on the first lecture for each course. Enrollment into the study was optional and no compensation or extra credit was provided. Students that consented to have their Moodle clickstream data also completed an optional demographic survey. This clickstream data was filtered to analyze the courses' lecture materials, which were hosted as PDF files with either "(light mode)" or "(dark mode)" appended to their filenames to indicate the color scheme of the file. The files were accessible for download at the start of the day for each lecture, adjacent to each other, and ordered with light-mode version appearing first. Students were presented with an end-of-semester survey asking for perceptions and attitudes toward the two color schemes. In total, 311 students consented to have their data analyzed for this study and 111 students completed the end-of-semester survey.

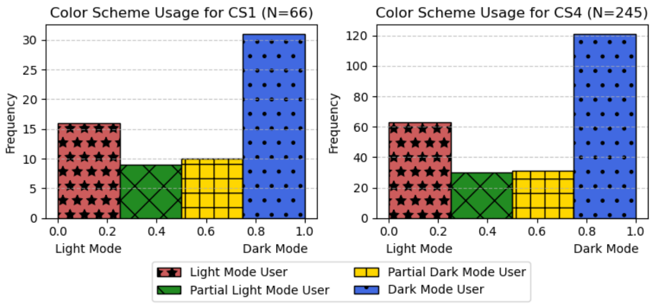


Figure 1: Dark Mode Prominence across Cohorts

4 Analysis

4.1 Color Scheme Prominence

We separate courses into two cohorts: CS1 to represent the two introductory courses ($N = 66$) and CS4 to represent the four upper level courses ($N = 245$). This is done to explore the difference between upper- and lower-level student preferences. We separated students into four quartiles based on their usage, referred as 'dark mode prominence' (DMP), shown in Figure 1. This calculation is a ratio of each student's selection of Dark Mode lecture materials and their total lecture material selections throughout the course. This ratio (Equation 1) was determined by the first PowerPoint downloaded by a student during a particular week, divided across the 16 week semester (e.g. a DMP score of 1.0 would mean the student always downloaded Dark Mode slides first).

$$\sum_{n=16}^{i} \frac{i_{first\_download}}{n}, \begin{cases} 1 & \text{if picked dark mode} \\ 0 & \text{if picked light mode} \end{cases} \tag{1}$$

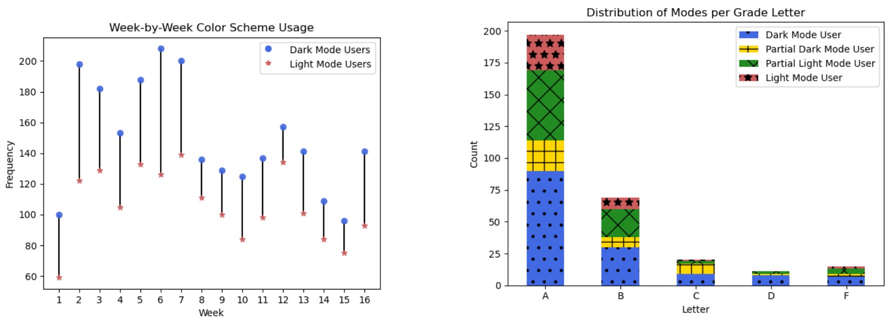
As shown in Figure 1, roughly half of students in both cohorts heavily favored the Dark Mode slides and a quarter favored Light Mode, while another quarter waned in their preference.

4.2 Color Scheme Selection Over Time

Since our findings showed similar distributions of color scheme preference in both cohorts, we combined the cohorts to look at all students throughout the 16 week lecture period. Figure 2a shows a consistent preference for Dark Mode lecture materials across the course. The increases around Week 12 and Week

16 reflect the courses' midterm exams. However, exams were not scheduled on the same weeks for CS1 and CS4 to maintain instructor's sanity.

4.3 Course Performance By Color Scheme Preference



(a) (b)

Figure 2: (a) Dark Mode Prominence Over Time; (b) Final Letter Grade by Color Scheme

While student primarily selected Dark Mode slides, we were not able to determine any significance on final course grades, $\chi^2(12, N = 311) = 18.33$, $p = 0.106$. However while the chi-squared results were not significant, Figure 2b shows a disproportionate representation of Dark Mode users also received an A in the course. A post-hoc analysis of chi-squared residuals showed no statistical deviations from this subgroup's expected frequency ($r = -0.097$). A lot of students made A's in the author's class because *reasons*.

5 Student Attitudes Toward Dark Mode

We presented students with a general comments survey at the end-of-the-semester survey which contained a single open-ended question asking students to describe their experiences and preferences regarding the dark mode and light mode lecture materials. Survey responses were given to ChatGPT for summarization, **which is one of the use cases for LLMs that people generally agree to be okay**[2]. We report that students responded to the survey considered the choice between color schemes to be a positive quality of life improvement ("*I liked having the option for both modes*"). Interestingly students perceived each mode supported their ability to focus on the lecture materials, with comments emphasizing the modes' clarity and contrast. However, our study was not design to appropriately evaluate these claims. Likewise, some comments included that dark mode improved students' '*ocular comfort*', with comments like "*helped with eye strain*", "*is much easier on my eyes*", and "*feels less harsh on the eyes*". Further some students reported their selection process was impacted by the time of day. Finally, some students reported they considered the options novel and purely cosmetic.

6 Discussion and Future Work

This research showed that while the majority of students preferred the Dark Mode version of lecture materials, color mode preference did not have a significant impact on students' course performance.

Future work will need to adopt 'time on task' and larger scale mass surveillance of student online mannerisms. Maybe incorporate some of TikTok's 'time on content' metrics to study what students are staring at and for how long? Need to also evaluate if they listen to Lofi Girl just to be on the safe side. Also should isolate participants' clothing to control for touch and additional methodologies to control for taste and smell (lavender is reported to be positively correlated to reducing anxiety, but may not taste as good as saltine crackers [7, 11]).

Instructors considering adopting a dark mode variant to their lecture materials may be concerned with the additional time required for their creation. Yeah. That's how time works. **BUT**, this did not take long for the author, so like, block out an additional hour or something.

If an instructor is creating both Light and Dark mode versions simultaneously, we recommend focusing on one mode initially before creating the second. Otherwise you spend way too much time fiddling. Pick one, finalize, **then fiddle**.

7 Limitations

There are several limitations toward this analysis.

1. The author was also the instructor. I have no idea if you'll have the same thing happen to your classes.

2. The color scheme was designed by people that have degrees in that kind of thing. Is it there fault the students' performance was not significant?

3. The students were CS students. They have to stare at screens more than other students, so maybe its just a CS thing?

4. Preferences ebb and flow like the tide. Maybe students will prefer Light Mode again in 5 to 50 years.

5. The researcher (also the author, also the instructor) didn't even wear a lab coat, therefore nullifying this entire experiment.

8 Call for Replication

The author encourages replication of this analysis with the reader's own cohorts and color schemes to encourage broader adoption. However as the author has seen from the various conferences this paper has been submitted to, it looks like reviewers have their own light/dark mode biases that interfere with their ability to appropriately review this work!

Like seriously, I've submitted this to 3 different venues and reviewers either **LOVE** this report or **DETEST** it. And then there's always the snide comment about the report not being *appropriate* for the venue. Fine, I'm gonna submit it to SIGBOVIK! **SURELY** that venue will respect and admire this work.

9 Conclusion

Instructors considering incorporating a Dark Mode variant to their course material should also review accent color options that provide sufficient contrast between the foreground and the background. While these color schemes were created by my institution's communications department, instructors could work with their own institutions or utilize free color palette resources, such as FlatUIColors.com. Instructors can reduce the time requirements for making these color schemes by utilizing Microsoft PowerPoint's Slide Master feature or similar features in other presentation software/applications.

tl;dr - color scheme does not impact student performance; however many students expressed support for the option of Dark Mode lecture materials in their survey responses. The irony of making this submission light mode is not lost to the author.

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